DONG (ADRIAN) AN

Data Engineer | Software Engineer | AI/ML Systems & HIPAA Data Governance
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PROFESSIONAL SUMMARY

Results-driven Data and Software Engineer with over 6 years of experience building and modernizing HIPAA-compliant data platforms for the California Department of Developmental Services. Led the Department's data de-identification guidelines (DDG) program by developing and implementing policies that enabled agency-wide compliance while reviewing and approving hundreds of Public Records Act requests. This initiative protected sensitive individual information and shielded the Department from potential liabilities, breaches, and litigation. Specializes in large-scale ETL/ELT migrations, successfully transitioning legacy on-premises SAS and Qlik pipelines to hybrid and cloud architectures on AWS Glue, Lambda, and Snowflake. These migrations reduced manual data production procedures through automations and enabled self-service data availability for public reporting and analytics. Developed and deployed a production-grade agentic AI system leveraging CrewAI for multi-agent orchestration, RAG architectures, AWS Bedrock for secure LLM inference, and a responsive React.js/Next.js frontend to automate de-identification quality reviews while maintaining strict HIPAA compliance and full auditability. Developed enterprise Power BI workflows and dashboards for both internal stakeholders and publicly released data products while training staff on self-service reporting through standardized models and governance guardrails. Applied an analytical approach combined with deep expertise in secure data governance and emerging AI/ML technologies to deliver scalable, automated, risk-reducing solutions.

TECHNICAL SKILLS

Data Platforms & Warehousing: Snowflake (Snowpipe, Streams, Tasks, secure views), AWS (Glue, Lambda, Step Functions, S3, Bedrock, SageMaker), Hybrid/CDC architectures, near-real-time pipelines

ETL/ELT & Data Integration: Python, SQL, SAS/Qlik legacy migrations, data lineage, validation frameworks, parallel cutover strategies, orchestration & schedulers

AI/ML & Agentic Systems: CrewAI (multi-agent orchestration), AWS Bedrock & SageMaker, RAG architectures, React.js/Next.js AI frontends, neural networks, AI/ML best practices

Business Intelligence & Administration: Power BI (enterprise + public dashboards, DAX, workflow automation), Qlik Data Catalog (on-premises administration & metadata), ArcGIS (administration & geospatial integration)

Data Governance, Privacy & Security: HIPAA Privacy Rule, California Information Practices Act (IPA) compliance, Safe Harbor & Expert Determination, PII/PHI risk assessment, allow-listing, encryption, suppression thresholds, Public Records Act request handling

Languages, Tools & Methods: Python, SQL, JavaScript/TypeScript, Git, ServiceNow, Snowflake SQL, Power BI DAX, Linux/Bash, cloud automation, SOP development, staff training, cross-functional collaboration (Legal, Programs, InfoSec)

PROFESSIONAL EXPERIENCE

California Department of Developmental Services (DDS) — IT Division, Data De-Identification Section | Sacramento, CA | July 2020 – Present

Software Engineer, Information Technology Specialist I | May 2025 – Present

- Architected and deployed a production-grade agentic AI system to automate quality assessment reviews of de-identified public data reports under the DDG program. Leveraged CrewAI for multi-agent orchestration, AWS Bedrock for secure LLM inference, and built a responsive React.js/Next.js frontend. This system streamlines manual review processes while maintaining strict HIPAA compliance and full auditability.
- Continue to lead technical aspects of the DDG program including final sign-off on data releases, risk assessments, and cross-agency coordination with 21 regional centers — translating complex compliance requirements into scalable, auditable data engineering solutions.

Data Engineer, Information Technology Specialist I | September 2020 – May 2025

- Led the Department's data de-identification guidelines (DDG) program, developing and implementing policies and processes that enabled department-wide privacy compliance. Reviewed and approved over 1000 DDG requests arising from legislative requirements for public information and Public Records Act

inquiries. This work has protected sensitive individual information and shielded the Department from potential liabilities, breaches, and litigation.

- Led the full-lifecycle migration of legacy on-premises SAS and Qlik ETL pipelines to a modern hybrid/cloud data platform on AWS Glue, Lambda, and Snowflake. Designed raw/staging/curated layering, implemented CDC patterns, and executed parallel validation during cutover — reducing manual processes by 80%.
- Served as on-premises administrator for Qlik Data Catalog, managing metadata, data lineage, and catalog governance. Additionally administered ArcGIS environments, supporting geospatial data integration critical to departmental operations and regional center coordination.
- Played a key role in developing and operationalizing the Department's enterprise Power BI workflow and dashboard ecosystem for both internal stakeholders and publicly available data products, improving data accessibility and self-service reporting through standardized models and governance guardrails.
- Designed and maintained secure data warehouse and BI solutions spanning on-premises data marts, AWS cloud platform, and hybrid architectures. Implemented comprehensive controls including allow-listing, encryption, role-based access, and rigorous data mapping/validation to ensure HIPAA and IPA compliance.
- Performed Expert Determination and re-identification risk assessments for de-identified data releases from 21 regional centers statewide. Developed SOPs and training materials, mentored staff on compliance workflows, and responsibility to provide technical support for court testimony on data governance and privacy practices.

Data Analyst, Information Technology Associate | July 2020 – September 2020

- Supported foundational data analysis, reporting, and monthly data production processes. Managed production data pipelines, refactored legacy code, and developed data marts for reporting workflows. Demonstrated strong technical aptitude leading to rapid promotion to IT Specialist I.

AWARDS & RECOGNITION

Superior Accomplishment Award — California Department of Developmental Services, December 2024

Recognized for outstanding leadership in DDG policy development, high-volume Public Records Act request handling, and protecting sensitive data and the Department from potential liabilities, breaches, and litigation.

EDUCATION

Master of Science, Computer Science — California State University, Sacramento (2019)

Focus: Distributed and Cloud Computing, Artificial Intelligence & Machine Learning

Bachelor of Science, Molecular, Cellular, and Developmental Biology — University of California, Santa Cruz

Training in the scientific method that supports careful, evidence-based, and detail-oriented work in software engineering and data governance.